Analytical chemistry 1 for engineers

Homework 2 – Fundamental knowledge

姓名: **Name:** **ID number:** **Major:**

H2-1. Write the name and number represented by each symbol. For example, for kW you should write kW = kilowatt = 10^3 watts. Note take care on the capital and lowercase.

(1) mmol (2) TJ (3) fg

(4) pm (5) μN (6) ns

(7) dPa (8) mL (9) MK

(10) GA

(1) mmol = millimole =10^{-3} mol (2) TJ = terajoule = 10^{12} J (3) fg= femtogram= 10^{-15} g
(4) pm = picometer = 10^{-12} m (5) μN = micronewton =10^{-6} N (6) ns= nanosecond=10^{-9} s
(7) dPa = decipascal = 10^{-1} Pa (8) mL= milliliter=10^{-3} L (9) MK= megakelvin=10^{6} K
(10) GA = gigaampere = 10^{9} A

H2-2. The concentration of the alkane $C_{20}H_{42}$ (M.W=282.56 g/mol) in a particular sample of rainwater is 0.2 ppb. Assume that the density of rainwater is close 1.00 g/mL and find the molarity of $C_{20}H_{42}$.

Due to the density of rainwater is close 1.00 g/mL,

$\Longrightarrow$ 0.2 ppb = 0.2 μg/L = 2×10^{-7} g/L;

FM($C_{20}H_{42}$) =282.56 g/mol;

$\Longrightarrow$ The molarity of $C_{20}H_{42}$ = $\frac{2\times10^{\wedge}(-7)\ g/L}{282.56\ g/mol}$ = 7×10^{-10} mol/L =0.7 nM

H2-3 The reaction $5H_2C_2O_4 + 2MnO_4^- + 6H^+ \rightarrow 10CO_2 + 2Mn^{2+} + 8H_2O$ requires 5 mol $H_2C_2O_4$ for 2 mol MnO_4^-. If you mix 1.15 g $Na_2C_2O_4$ (M.W 134.00 g/mol) with 0.60 g $KMnO_4$ (M.W 158.03 g/mol) and excess aqueous acid. Which reactant is limiting? How much CO_2 should be produced?

mole of $Na_2C_2O_4$: $\frac{1.15g}{134.00\ g/mol}$ =8.58 mmol;

mole of $KMnO_4$: $\frac{0.60\ g}{158.03\ g/mol}$ =3.8 mmol;

	$5Na_2C_2O_4$	$+2MnO_4^-$	$+ 6H^+$	$\rightarrow 10CO_2$	$+ 2Mn^{2+}$
Initial (mmol)	8.58	3.8	excess	0	0

8.58 mmol $Na_2C_2O_4$ requires 8.58 mmol $\times \frac{2}{5}$ $KMnO_4$ = 3.432 mmol $KMnO_4$;

And, there provides 3.8 mmol $KMnO_4$, which is enough to react with $Na_2C_2O_4$.

So, $Na_2C_2O_4$ is the limiting reactant, and 17.2 mmol of CO_2 is produced.

	$5Na_2C_2O_4$	$+2MnO_4^-$	$+ 6H^+$	$\rightarrow 10CO_2$	$+ 2Mn^{2+}$
Initial (mmol)	8.58	3.8	excess	0	0
Final (mmol)	0	0.37	many	17.2	3.43